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SUPERSEDING
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21 December 1959

MILITARY SPECIFICATION

CARTRIDGE, 106MM, DRILL (INERT), M368, FOR RIFLE, M40A1 AND M40A1C

1. SCOPE

1.1 This specification covers Cartridge, 106MM, Drill (Inert), M368, for Rifle M40A1 and M40A1C.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550 - Ammunition and Special Weapons, General Specification For

STANDARDS

Federal

Federal Test Method Standard No. 141 - Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling and Testing

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedure and Tables for Inspection by Attributes

FSC 1315

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2.1 (Cont'd)

DRAWINGS

Munitions Command

F8596153	- Cartridge, Drill (Inert) M368, for 106MM M40A1 and M40A1C Rifle, Assembly and Marking Diagram
F7549071	- Container, Ammunition, Fiber, M314 for 106MM Rifle Ammunition
F7549070	- Box, Packing, Ammunition, for 106MM Rifle Ammunition in Fiber Container M314
D8596154	- Loading Assembly (Inert)
IEL-8596153	- Inspection Equipment List for Cartridge, Drill (Inert) for 106MM, M40A1 and M40A1C Rifle

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Material.-The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture or assembly.

3.2.1 Government furnished materials.-When the Government elects to supply one or more components of the cartridge assembly, it will be the responsibility of the assembling contractor to ascertain that the component complies with the requirements of the assembly drawing.

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3.2.2 Loading assembly.-Components of the loading assembly not supplied by the Government shall comply with the requirements of the applicable metal parts assembly drawing.

3.3 Protective coating.-The protective coating shall be in accordance with the applicable drawings and the specifications referenced thereon.

3.4 Workmanship.-The requirements for workmanship are as shown on the applicable drawings, the referenced specifications, and the following.

3.4.1 Metal defects.-The assembly shall be free of cracks, splits, dents and other similar defects.

3.4.2 Threads.-Threads shall be fully formed and undamaged for the entire minimum length and depth as specified on the applicable drawings and in MIL-A-2550.

3.4.3 Foreign matter.-No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter.

3.5 Inert filler.-The constituents of the cast inert filler shall be varied in the proportion necessary to produce the loaded shell weight requirement of Drawing D8596154.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

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4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of the following:

3 complete assemblies, painted
3 loading assemblies (inert)
3 metal parts assemblies (unpainted)
3 metal parts assemblies (painted)
3 case and liner assemblies
3 dummy primers
3 connecting rods
3 cartridge cases (uncoated)
3 bodies
3 base plugs
3 gaskets

The initial production sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specification shall result in sample disapproval. The Government reserves the right to require new initial production samples until an acceptable sample is submitted.

4.3 Inspection.

4.3.1 Government furnished materials.-When the Government elects to supply one or more components, the contractor shall perform only such visual examination of the components as deemed necessary to assure that the assembly contains no component that may have been damaged in storage or shipment to such an extent that an assembly containing such a component would be rendered useless for its intended use.

4.3.2 Lot.

4.3.2.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.3 Examination.-Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination.

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4.3.3.1 Plug Base, Drawing C8596148

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Concentricity of pitch diameter of body thread with pitch diameter of fuze thread, including perpendicularity with body seat and fuze seat	Gage
202. Perpendicularity of body seat with pitch diameter of body thread	Gage
203. Pitch diameter of body thread, min.	Gage
204. Pitch diameter of body thread, max.	Gage
205. Major diameter of body thread, min.	Gage
206. Pitch diameter of fuze thread, max.	Gage
207. Pitch diameter of fuze thread, min.	Gage
208. Minor diameter of fuze thread, max.	Gage
209. Diameter of body thread chamfer, min.	Gage
210. Diameter of fuze thread chamfer, max.	Gage
211. Diameter of gasket boss	Gage
212. Diameter of flange	Gage
213. Diameter of flat between body thread and gasket boss	Gage
214. Length, total	Gage
215. Thickness of flange	Gage
216. Width of gasket boss	Gage
217. Width from end of body thread to start of gasket boss	Gage
218. Metal defective	Visual
219. Presence and location of wrench holes	Visual (See 6.2)
220. Radii or chamfers missing or incorrect	Visual (See 6.2)
221. Surface finish improper	Visual (See 6.2)
222. Protective coating inadequate or damaged	Visual (See 6.2)
223. Burr	Visual
224. Foreign matter	Visual

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4.3.3.2 Rod, Connecting, Drawing D8596157

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Pitch diameter of external thread, min.	Gage
202.	Pitch diameter of external thread, max.	Gage
203.	Major diameter of external thread, min.	Gage
204.	Pitch diameter of internal thread, max.	Gage
205.	Pitch diameter of internal thread, min.	Gage
206.	Minor diameter of internal thread, max.	Gage
207.	Depth of cavity	Gage
208.	Length of internal effective thread, min.	Gage
209.	Length of external effective thread, min.	Gage
210.	Length, total	Gage
211.	Metal defective	Visual
212.	Surface finish, improper	Visual (See 6.2)
213.	Burr	Visual
214.	Radii or chamfers missing or incorrect	Visual (See 6.2)
215.	Foreign matter	Visual

4.3.3.3 Primer, Dummy, Drawing C8596158

CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Concentricity of body with pitch diameter of thread	Gage
202.	Pitch diameter of thread, min.	Gage
203.	Pitch diameter of thread, max.	Gage

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4.3.3.3 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
204. Major diameter of thread, min.	Gage
205. Diameter of body between thread and head	Gage
206. Diameter of head	Gage
207. Thickness of head	Gage
208. Length of effective thread, min.	Gage
209. Length total	Gage
210. Presence and location of wrench holes	Visual (See 6.2)
211. Radii or chamfers missing or incorrect	Visual (See 6.2)
212. Surface finish improper	Visual (See 6.2)
213. Protective coating inadequate or damaged	Visual (See 6.2)
214. Burr	Visual
215. Metal defective	Visual
216. Foreign matter	Visual

4.3.3.4 Gasket, Drawing B8596149

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Hardness	Approved Method
202. Diameter, inside	Gage
203. Diameter, outside	Gage
204. Thickness	Gage

4.3.3.5 Body, Drawing F8596146

CRITICAL

None defined

MAJOR

101. Location of index button hole	Gage

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4.3.3.5 (Cont'd)

	Defect	Method of Inspection
MINOR		
201.	Wall thickness at index button hole, min.	Gage
202.	Metal defective	Visual
203.	Radii or chamfers, missing or incorrect	Visual (See 6.2)
204.	Surface finish improper	Visual (See 6.2)
205.	Burr	Visual
206.	Foreign matter	Visual

4.3.3.6 Metal Parts Assembly, Drawing F8596145 (before paint)

CRITICAL		
	None defined	
MAJOR		
101.	Rotating band groove functionally concentric with rotating band lands	Gage
102.	Alignment of index button(s) with rotating band lands	Gage
103.	Diameter of rotating band lands	Gage
104.	Diameter of rotating band grooves	Gage
105.	Height of index button	Gage
106.	Width of rotating band lands	Gage
MINOR		
201.	Concentricity exterior of nose with bourrelet	Gage
202.	Concentricity interior of nose with bourrelet	Gage
203.	Concentricity of body with bourrelet	Gage
204.	Concentricity of pitch diameter of thread with bourrelet, including perpendicularity with rear face	Gage
205.	Rotating band lands functionally concentric with bourrelet	Gage
206.	Diameter of bourrelet, min.	Gage
207.	Diameter from rear face to first inside basic length	Gage
208.	Diameter from rear face to second inside basic length	Gage
209.	Diameter from rear face to fourth inside basic length	Gage

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4.3.3.6 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
210. Length from nose to first outside basic diameter	Gage
211. Length from nose to second outside basic diameter	Gage
212. Length of rotating band	Gage
213. Length, total	Gage
214. Location of rotating band from rear face	Gage
215. Location of bourrelet from front of rotating band, max.	Gage
216. Location of crimp groove rib	Gage
217. Wall thickness, min. (indicated area)	Gage
218. Wall thickness at nose	Gage
219. Width of crimp relief groove	Gage
220. Width of crimp groove rib	Gage
221. Width of groove forward of rib	Gage
222. Radius rear of crimp relief groove	Gage
223. Diameter of crimp relief groove	Gage
224. Diameter of body front and rear of rotating band	Gage
225. Diameter forward of thread	Gage
226. Diameter of chamfer at rear of thread	Gage
227. Length from rear of rotating band to rear of angle	Gage
228. Depth of undercut front and rear of rotating band (if applicable)	Visual (See 6.2)
229. Width of undercut front and rear of rotating band (if applicable)	Visual (See 6.2)
230. Radii or chamfers missing or incorrect	Visual (See 6.2)
231. Surface finish, improper	Visual (See 6.2)
232. Metal defective	Visual
233. Burr	Visual
234. Foreign matter	Visual

4.3.3.7 Metal Parts Assembly, Drawing F8596145 (after paint)

CRITICAL

None defined

MAJOR

None defined

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4.3.3.7 (Cont'd)

Defect		Method of Inspection
MINOR		
201.	Weight	Approved Scale
202.	Pitch diameter of thread, max.	Gage
203.	Pitch diameter of thread, min.	Gage
204.	Minor diameter of thread, max.	Gage
205.	Length of effective thread, min.	Gage
206.	Length of effective thread, max.	Gage
207.	Diameter of bourrelet, max.	Gage
208.	Paint in cavity, inadequate or excessive	Visual
209.	Improper assembly	Visual and Manual
210.	Rotating band damaged	Visual
211.	Paint on rotating band bearing surfaces and rear face	Visual
212.	Paint on threads	Visual
213.	Paint inadequate or damaged	Visual
214.	Paint not dry	Visual and Tactile
215.	Marking missing, incorrect or illegible	Visual
216.	Burr	Visual
217.	Foreign matter	Visual

4.3.3.8 Loading Assembly (Inert), Drawing D8596154

CRITICAL

None defined

MAJOR

None defined

MINOR		
201.	Weight	Approved Scale
202.	Button(s) missing or damaged	Visual
203.	Rotating band damaged	Visual
204.	Base plug not properly assembled	Visual
205.	Body dented or damaged	Visual
206.	Gasket missing between base plug and body	Visual
207.	Depth to inert filler (before assembly of base plug)	Gage

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4.3.3.9 Case, Cartridge, Drawing F8596150 (prior to coating)

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Inside diameter of mouth functionally concentric with body	Gage
202.	Primer hole counterbore functionally concentric with primer hole	Gage
203.	Primer hole functionally concentric with body	Gage
204.	Flange diameter functionally concentric with body	Gage
205.	Diameter of shoulder under flange concentric with body	Gage
206.	Diameter of mouth, inside	Gage
207.	Diameter of flange, min.	Gage
208.	Diameter of primer hole	Gage
209.	Taper in primer hole incorrect	Gage
210.	Thickness of flange, min.	Gage
211.	Thickness of wall at mouth, min.	Gage
212.	Thickness of wall at min. length	Gage
213.	Thickness of wall near head	Gage
214.	Depth of primer hole counterbore, min.	Gage
215.	Length from shoulder forward of flange to bottom of primer hole counterbore, min.	Gage
216.	Diameter of primer hole counterbore	Gage
217.	Diameter of body under shoulder, min.	Gage
218.	Diameter of shoulder under flange, min.	Gage
219.	Diameter of undercut under shoulder, min. (if applicable)	Gage
220.	Length, total	Gage
221.	Width of undercut, max. (if applicable)	Gage
222.	Thickness through head, min.	Gage
223.	Location of first row circumferential holes	Gage
224.	Spacing and alignment of holes, improper	Visual
225.	Row of holes missing	Visual
226.	Welding extended into perforated area	Visual
227.	Weld not flush with inside diameter of mouth for indicated length	Visual
228.	Metal defective	Visual

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4.3.3.9 (Cont'd)

	Defect	Method of Inspection
MINOR (Cont'd)		
229.	Surface finish improper	Visual (See 6.2)
230.	Chamfer missing on head or incorrect	Visual (See 6.2)
231.	Marking missing, incorrect or illegible	Visual
232.	Burr	Visual
233.	Foreign matter	Visual

4.3.3.10 Case, Cartridge, Drawing F8596150 (after coating)

CRITICAL

None defined

MAJOR

101.	Maximum profile	Gage

MINOR

201.	Length, min. of inside diameter at mouth	Gage
202.	Case dented	Visual
203.	Protective coating inadequate or damaged	Visual
204.	Varnish in primer hole	Visual
205.	Compound missing from primer hole	Visual
206.	Foreign matter	Visual

4.3.3.11 Case and Liner Assembly, Drawing F8596155

CRITICAL

None defined

MAJOR

None defined

MINOR

201.	Diameter of mouth and length to dimples	Gage
202.	Diameter at dimples	Gage
203.	Location of dimples	Gage
204.	Case dented or damaged	Visual
205.	Liner loose or damaged	Visual
206.	Protective coating inadequate or damaged	Visual

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4.3.3.12 Cartridge, Assembly, Drawing F8596153

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Insecure assembly (torque test)	Approved Device
102. Dummy primer extends beyond face of case base	Visual and Tactile
MINOR	
201. Length, total, max.	Gage
202. Diameter of projectile after painting, max.	Gage
203. Protective coating inadequate or damaged	Visual (See 6.2)
204. Marking missing, incorrect or illegible	Visual
205. Burr	Visual
206. Foreign matter	Visual

4.3.3.13 Container, Sealed Fiber, Drawing F7549071

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Length of strip, sealing, improper	Gage
202. Contents loose	Manual
203. Gap between cover and body excessive	Visual
204. Tapping incomplete or badly wrinkled	Visual
205. Color of tape incorrect	Visual
206. Cuts or gouges in outer layer	Visual
207. Metal ends loose or distorted	Visual
208. Marking missing, incorrect or illegible	Visual
209. Container cut or damaged through all layers	Visual

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4.3.3.14 Box, Wood, Sealed, Drawing C7549070

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Screw protruding outside box	Visual
202.	Hardware missing, broken or insecure	Manual and Visual
203.	Contents loose	Manual
204.	Marking missing, incorrect or illegible	Visual
205.	Car seal missing, unsealed or improperly assembled	Manual and Visual
206.	Box damaged	Visual

4.3.4 Chambering.-Each round shall be chambered in a 106MM, M40 rifle or a government certified profile and alignment gage. Failure of any round to chamber freely shall be cause for rejection of the failing round.

4.3.5 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.5.1 Cleaning and application of protective coating.-The sample size and acceptance criteria for the inspection of the cleaning of the item, application of protective coating, and salt spray testing shall be as specified in the applicable drawing and specification. Failure to comply with the requirements shall be cause for taking the corrective action recommended therein.

4.3.6 Inspection equipment.-Index of Inspection Equipment List Number IEL-8663876 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with the code designation specified on the applicable IEL.

4.4 Test methods and procedures.

4.4.1 Protective coating.-The procedures for testing the cleaning and surface treatment methods for the components and assemblies shall be in accordance with the methods and criteria for testing as specified in the applicable specification. The salt spray test prescribed by the applicable drawing shall be performed utilizing the equipment and procedures specified by Federal Test Method Standard No. 141.

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5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A.-Packing and marking shall be in accordance with drawings F7549070 and F7549071, except that the nomenclature shall be in accordance with the assembly drawing.

5.2 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Loading assembly.-The metal parts assembly referenced is noted on the loading assembly (inert) drawing.

6.2 Visual examination qualification.-When compliance with the applicable requirements is in doubt, as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.3 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.4 This specification includes the requirements of Purchase Description FA-PD-MI-2380, Rev. 0 dated 21 December 1959.

6.5 International agreements.-Certain provisions of this Specification are the subject of ABCA-Army-Standard-46C. The international agreements concerned will be considered in formulating any proposal for amendment, revision or cancellation of this Specification. When the departmental custodians propose to make any change which might affect compatibility of the item with international standardization agreements, they will so inform their respective Departmental Standardization Office so that appropriate action may be taken respecting the international agreements concerned.

Custodian:

Army - MU

Preparing activity:

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